

Youdahe Asfaw

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EDUCATION

Minnesota State University Mankato

Bachelor of Science in Computer Science & Statistics

Expected Graduation - June 2028

3.5/4.0

TECHNICAL SKILLS

Languages: Python, TypeScript, Go, Rust, Java, JavaScript, SQL

Frameworks: FastAPI, React, LangChain/LangGraph, Express

Developer Tools: Docker, Kubernetes, AWS, PostgreSQL, Redis, Qdrant, Prometheus/Grafana, Git

EXPERIENCE

Software Engineer Intern

NMDP

Aug 2026 – Nov 2026

Minneapolis, MN

- Joining the data platform team supporting donor-patient matching infrastructure.

Software Engineer Intern

Liberty Mutual Insurance

May 2026 – Aug 2026

Portsmouth, NH

- Built an internal self-service application automating AIX node retirement for the REX platform team, cutting time to decommission from 2–3 weeks to 1 day by replacing manual sign-off from up to 20 engineers with real-time connection monitoring; used to retire 15+ nodes.
- Supported migration of on-prem workloads to Kubernetes using StatefulSets, PVC-backed storage, and CNI-configured pod networking.

Software Engineer Intern

Medica

Jan 2026 – May 2026

Minnetonka, MN

- Built and shipped to production a document ingestion pipeline over 50K+ internal documents, covering chunking, embedding generation, and vector indexing with metadata filtering, achieving 95% precision and recall on a held-out 100+ conversation LLM-as-a-judge eval set.
- Developed a hybrid sparse-dense retrieval and reranking layer for a stateful onboarding assistant handling document Q&A, page routing, and provider support; reduced onboarding task time from 1 week to 2 days in QA testing.

Machine Learning Research Assistant

Gustavus Adolphus College

Feb 2025 – Jun 2025

Saint Peter, MN

- Built a document intelligence pipeline in Python using Azure Form Recognizer and LayoutLMv3 to extract structured fields from I-9s, W-2s, and tax-exempt forms, cutting end-to-end processing time 75% (60 to 15 minutes) in A/B testing against the existing manual workflow.
- Developed language-agnostic extraction using layout, field positioning, and visual structure rather than keyword rules, generalizing across digital forms, photos, uploads, and scans at 85% field accuracy with high-80s precision and recall; presented at four national symposia including Johns Hopkins Macksey and UChicago PRISM.

Software Engineer Intern

Medtronic

Aug 2023 – Aug 2024

Mounds View, MN

- Built core ingestion components for a centralized data platform consolidating 500K+ entries from fragmented legacy sources (Excel inventories, YAML configuration records, device telemetry, event logs, MP4 diagnostic recordings), designing a stateless, modular ingestion layer that normalized heterogeneous formats without coupling data availability to individual team services.
- Added record-level validation and logging to surface failed and incomplete ingestions automatically, replacing manual line-by-line review across spreadsheet inventories and legacy sources.

PROJECTS

Docere | *Python, TypeScript, FastAPI, PostgreSQL, Redis, Docker, Kubernetes*

- Built an AI tutoring platform that adapts to student knowledge gaps, growing to around 500 active users and a signed contract covering 20,000+ members; raised \$25K through grants and pitch competitions.
- Engineered Qdrant vector retrieval with adaptive strategy selection, an async pipeline (FastAPI, Redis/ARQ, LangGraph), and Canvas/Moodle integrations via LTI 1.3.

Distributed Event Streaming Platform | *Go, Linux*

- Built a Kafka-style distributed commit log in Go with partitioned, replicated storage and tiered data migration (memory-mapped to SSD to object storage), instrumented with Prometheus/Grafana for per-partition throughput, latency percentiles, and consumer lag.
- Implemented exactly-once delivery via idempotent producers and transactional offset commits, plus stream processing with event-time windowing and checkpoint/restore recovery over a custom binary TCP protocol.